^{1}H DOSY NMR Determination of the Molecular Weight and the Solution Properties of Poly(*N*-acryloylmorpholine) in Various Solvents

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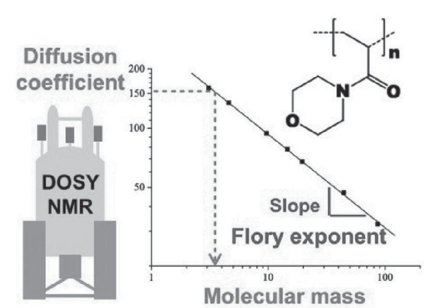
^{1}H DOSY NMR provides a simple and convenient method to determine both the molecular weight and the solution properties of poly(*N*-acryloylmorpholine) (PNAM) in various organic and aqueous solvents. This technique is used to measure the average diffusion coefficient D_{PNAM} of a library of PNAM samples of increasing molecular weights synthesized by reversible addition fragmentation chain transfer controlled radical polymerization. In all solvents, the highly linear relationship between Log D_{PNAM} and Log M_w yields calibration curves that can be used for the reliable determination of the molecular weight of various types of PNAM samples. In addition, the slope of the calibration curves can be considered as the Flory exponent δ characterizing the solvent–polymer interaction. The measurements show that $CDCl_3$ and D_2O are theta solvents for PNAM and that δ slightly decreases in phosphate buffer solution (PBS), pH 7.4, and upon addition of MeOD in D_2O.



1. Introduction

N-acryloylmorpholine (NAM) is a bi-substituted acrylamide derivative exhibiting several outstanding features such as good solubility in a wide range of aqueous and organic solvents and the ability to yield high molecular weight polymers that are also soluble both in aqueous and in polar or low polarity solvents.[1–5] Poly(*N*-acryloylmorpholine) (PNAM; Figure 1) is an attractive alternative to polyethyleneglycol (PEG) since both polymers are highly biocompatible[6] and show similar properties.[7] In addition, PNAM is synthesized by radical polymerization (unlike PEG). Therefore, NAM can be advantageously copolymerized with a large choice of co-monomers such as functional monomers that can be used to bind biomolecules or other entities of interest in lateral position of the polymer chains.

For many years now, PNAM and related copolymers have been considered for various bio-related applications. Cross-linked networks were used for gel-phase synthesis of peptides,[8–10] semi-permeable membranes for plasma separation,[11] polymeric supports for gel chromatography[12,13] and capillary electrophoresis.[14] Linear PNAM was also used for drug delivery applications, protecting liposomes from blood clearance,[15] or increasing the lifetime of enzymes both in vitro and in vivo.[16]

More recently, we and others conducted in-depth studies about the homo-[17–20] and co-polymerization[21,22] of NAM by the reversible addition fragmentation chain transfer (RAFT) controlled radical polymerization

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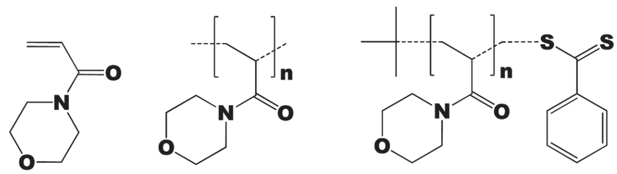

Figure 1. *N*-acryloylmorpholine monomer (NAM) and poly(*N*-acryloylmorpholine) (PNAM) homopolymer. *t*Bu-PNAM-DB represents the PNAM homopolymers synthesized by RAFT polymerization with a *tert*-butyl (*t*Bu) α-chain end and a dithiobenzoate (DB) ω-chain end.

technique that was introduced by Rizzardo and co-workers at CSIRO, Australia, in 1998.[23] Well-defined PNAM-based polymers were obtained with predictable molecular weights, narrow molecular weight distributions, and controlled chain-end functionalities as evidenced by size exclusion chromatography coupled with multiangle laser light scattering detection (SEC-MALLS), proton nuclear magnetic resonance (¹H NMR), and MALDI-TOF mass spectrometry (MALDI-TOF-MS).[24] Low dispersity samples ($Đ$ < 1.2) were obtained with a number-average molecular weight up to 300 000 g mol^{-1}.[25] In addition, various polymer architectures and self-assemblies were prepared such as block copolymers, nanoparticles, liposomes, and Lipoparticles.[21,26–29] Statistical copolymerization of NAM especially with *N*-acryloxysuccinimide, a monomer bearing an activated ester function, yielded versatile reactive polymer precursors that can be used to elaborate multifunctional polymers via post-polymerization functionalization. This strategy was taken into profit to prepare different (bio)conjugates for specific biological applications.[22,30–33]

Yet, very little is known about the solution properties of PNAM, especially in aqueous media. We thus thought that the recent advances in Diffusion-Ordered NMR SpectroscopY (DOSY NMR) of polymers could help us to gain more insights.

Pulse Field-Gradient Spin Echo NMR experiments were first described by Stejskal and Tanner to study the diffusion of molecular species in solution.[34] Then, Morris and Johnson[35] introduced the 2D approach, called DOSY, to correlate the translational diffusion coefficient with the chemical shifts. DOSY NMR was initially designed for the analysis and the characterization of mixtures. Johnson was the first to show that it could also be used to determine the molecular weight distributions of PEG samples.[36]

Walderhaug et al. recently reviewed various studies performed by field-gradient NMR on polymer systems including polymer solutions.[37] DOSY NMR has been proposed as an efficient method to study homopolymer mixtures,[38] to reveal the colloid-like behavior of a diblock copolymer,[39] and to elucidate the structure of block copolymers.[40] These studies indicated that DOSY NMR is a powerful tool to evaluate the efficiency of purification processes by detecting traces of impurities in (co)polymer samples, such as unreacted monomers, residual homopolymers, or degradation products.

In 2009, Delsuc and co-workers showed that NMR measurements of the diffusion coefficient are relevant to determine the fractal dimension of objects, d_F, that describes the way they are filling the space in 3D.[41] They confirmed that, for diluted solutions of linear polymers, the exponent α of the power law correlating the diffusion coefficient D and the molecular mass M (Equation (1)) could be identified as the Flory exponent δ correlating the gyration radius and M (Equation (2)). The Flory exponent is usually considered as the inverse of d_F (Equation (3)) and is indicative of the polymer–solvent interaction strength. δ is predicted to be ⅓, 0.5 and 0.588, respectively, for a poor solvent, a θ solvent and a good solvent of the polymer

$$D \propto M^{-\alpha} \tag{1}$$

$$R_g \propto M^{-\delta} \tag{2}$$

$$d_F = 1/\delta \tag{3}$$

DOSY NMR has also been used to determine the molecular weight of different polymers. Viel et al. estimated the molar mass and the hydrodynamic radius (R_h) of uncharged polysaccharides[42] and poly(amidoamine) dendrimers.[43] This method has then been extended to the determination of the molar mass of block copolymers[44] and oligosaccharides[45] by calibrating the diffusion dimension with structurally similar polymer standards. In addition, a DOSY procedure was set up by Delsuc and co-workers to access the dispersity $Đ$ of linear polymers.[46] In 2012, Grubbs and co-workers proposed to use DOSY NMR to determine the weight-average molecular weight M_w for kinetic samples of linear and narrowly distributed polymers produced from various living/controlled polymerization processes.[47]

In this study, we investigated the applicability of such ¹H DOSY NMR measurements to PNAM samples synthesized by RAFT polymerization. Then, ¹H DOSY NMR was used to estimate the Flory exponents for PNAM in different aqueous and organic solvents. Calibration curves were established to be used for the reliable determination of the weight-average molecular weight of various types of PNAM samples.

2. Experimental Section

2.1. Materials

NAM (Aldrich, 97%) was distilled under reduced pressure (120 °C; 10 mm Hg) to remove the inhibitor. *Tert*-Butyl dithiobenzoate (*t*BDB) was synthesized by a one-step process resulting in very high yield and purity (>98%).[17] Azobisisobutyronitrile was purchased from Aldrich (>99%) and purified by crystallization from ethanol. 1,4-Dioxane (Carlo erba, 99%) was distilled over $LiAlH_4$ and stored at 4 °C under nitrogen. Diethyl ether (SDS, 99.5%) was used as received. All NMR solvents D_2O, $CDCl_3$, and MeOD were purchased from Eurisotop and used as received. Deuterated phosphate buffer (PBS, pH 7.4) was prepared according to Hage and co-workers.[48]

2.2. NAM Polymerizations

PNAM homopolymers were synthesized by RAFT polymerization in Schlenk tubes or in a 12-vial parallel reactor (Radleys Carousel 12 Plus Reaction Station) following a previously optimized RAFT polymerization procedure.[18] PNAM samples were purified by two consecutive precipitations in a large volume of diethyl ether before drying under vacuum. Polymer molecular weights and molecular weight distributions were obtained by size exclusion chromatography coupled with a three-angle laser light scattering detector (Wyatt Technologies) (SEC/MALLS), ^{1}H NMR, and MALDI-TOF-MS (see below).

2.3. Analytical Techniques

2.3.1. ^{1}H NMR

Regular ^{1}H NMR experiments were conducted in deuterated chloroform at 298 K on a Bruker Ultrashield spectrometer operating at 300.13 MHz and on a Bruker 500 Ultra Shield spectrometer operating at 500.1 MHz.

2.3.2. SEC/MALLS

SEC/MALLS was performed using a setup composed of a Shimadzu LC-6A liquid chromatography pump and a PLgel Mixed-C column (5 µm size pores). Online double detection was provided by a differential refractometer (DRI Waters 410) and a three-angle (46°, 90°, 133°) MiniDAWN TREOS light scattering photometer (Wyatt Technologies), operating at 658 nm. Analyses were performed by injection of 70 µL of polymer solution (5 mg mL^{-1}) in chloroform. The specific refractive index increment (dn/dc) for PNAM in the same eluent (0.130 mL g^{-1}) was previously determined with an NFT ScanRef monocolor interferometer operating at 633 nm. The molar mass and dispersity data were determined using the Wyatt ASTRA SEC/ LS software package.

2.3.3. Matrix-Assisted Laser Desorption Ionization Time-Of-Flight Mass Spectrometer (MALDI-TOF-MS)

MALDI-TOF-MS was conducted on a Voyager-DE STR (Applied Biosystems, Foster City, CA). This instrument was equipped with a nitrogen laser (wavelength 337 nm) to desorb and to ionize the samples. The accelerating voltage used was 20 or 25 kV. The positive ions were detected in all cases. The spectra were the sum of 200 shots, and an external mass calibration was used (Sequazyme peptide mixture). This MALDI instrument used a stainless steel target, on which the samples were deposited and dried. Samples were prepared by dissolving the polymer in THF at a concentration of 10 g L^{-1}. The matrix was 3-indoleacrylic acid (Fluka, Milwaukee, WI), used without further purification and dissolved in dimethylformamide (DMF; 0.25 M). Matrix and polymer solutions were mixed at 1/1 vol/vol ratio, and then 0.5 µL of the mixture was deposited onto the MALDI target before insertion into the ion source chamber.

2.3.4. ^{1}H DOSY NMR

All DOSY NMR experiments were carried out at 300 K on a Bruker Avance III spectrometer operating at 400.13 MHz for ^{1}H with a 5 mm BBFO+ probe equipped with z-gradient coil delivering up to 52.8 G cm^{-1}. Gradient coil strength was calibrated with a 'doped water' NMR reference standard ($GdCl_3$ 0.1 mg mL^{-1} in D_2O with 1% H_2O and 0.1% CH_3OH-^{13}C, CortecNet). Temperature was calibrated with an 80 wt% ethylene glycol solution in DMSO-d$_6$ (NMR reference standard) and was regulated to avoid fluctuations due to sample heating. Samples were prepared by dissolving the weighted polymer powder in 800 µL of deuterated solvent. Prior to each experiment, the NMR tubes were inserted 15 min in the spectrometer for temperature equilibration. DOSY experiments were conducted with a ceramic spinner without rotation to avoid convection. Proton 90° pulse was measured.

DOSY experiments in D_2O were acquired using the Bruker LEDBPGPPR2S[49] pulse program (stimulated echo sequence with bipolar gradient pulses) including a pre-saturation block during relaxation delay D1 and diffusion time Δ to suppress the water signal. The gradient pulse length δ was optimized for each sample and varied between 1800 and 4000 µs. The diffusion delay was set to 100 ms. Due to its lower viscosity, chloroform is more sensitive than D_2O to temperature fluctuations. DOSY experiments in $CDCl_3$ were acquired using the Bruker DSTEBPGP3S pulse program (double stimulated echo sequence with bipolar gradient pulses) to remove the effect of constant flows in the sample due to thermal convection.[38] The gradient pulse length δ was optimized for each sample and varied between 1300 and 3000 µs. The diffusion delay was set to 100 ms. For all experiments, gradients were linearly sampled from 0.476 G cm^{-1} to 47.115 G cm^{-1} in 40 points and the gradient pulses were smooth-square shaped (SMSQ10.100). Sixty four scans were acquired on 16k data points (1 s recycle time). Fourier transform was applied to the free induction decays (FIDs) with 2 Hz Lorentzian line-broadening using the Topspin software (Bruker) and an automatic baseline correction routine was applied. Manual baseline refinement had to be conducted in some cases. For all pulse sequences, the experimental signal amplitude (I) of an NMR resonance is given by the Stejskal–Tanner equation[34]

$$I = I_o\, e^{-D\gamma^2 g^2 \delta^2 (\Delta - \delta/3 - \tau/2)} \tag{4}$$

where I_0 is the reference amplitude (at zero gradient strength), D is the diffusion coefficient, γ is the gyromagnetic ratio of the observed nucleus, τ is the time between the bipolar gradient pulses, g is the gradient strength, δ is the gradient pulse length, and Δ is the diffusion time (self-diffusion).

From the 2D-DOSY spectra, the diffusion coefficients were estimated by Inverse Laplace[50] using the DOSYm software.[51] The processing was computed on 256 points. This method uses a maximum entropy algorithm which is able to distinguish between the different compounds of a complex mixture with overlapping signals. Furthermore, no hypothesis on the number and the size of constituents is required.

3. Results and Discussion

3.1. Synthesis of PNAM Standard Samples with Increasing Molecular Weights

Various PNAM homopolymer samples with increasing molecular weights (MW) and low dispersity *Đ* were synthesized by RAFT polymerization using *tert*-butyl dithiobenzoate as control agent (Table 1). The synthesis was carried out in parallel reactors or in Schlenk tubes following previously described procedures (see the Experimental Section). The absolute MW distribution, M_n, M_w, and *Đ* of the resulting *t*Bu-PNAM-DB samples bearing a *tert*-butyl α-chain end and a dithiobenzoate ω-chain end (Figure 1) were characterized after purification by SEC/MALLS (Table 1). Complementary data were also acquired from ^{1}H NMR and MALDI-TOF-MS, only for low MW samples. M_n values were calculated from ^{1}H NMR spectra by comparison of the integrals corresponding to the protons of the NAM repetitive units with the integrals corresponding to the chain-end protons. MALDI-TOF-MS spectrogram also gave access to M_n, M_w, and *Đ* values. As previously reported, these results were in very good agreement with those obtained by SEC/MALLS.[24]

Table 1. Characteristics of the molecular weight distribution of the *t*Bu-PNAM-DB samples synthesized by RAFT polymerization, determined by SEC/MALLS, ^{1}H NMR, and MALDI-TOF-MS.

Sample	SEC/MALLS		
	M_n [g mol^{-1}]	M_w [g mol^{-1}]	*Đ*
P1[a)]	3300	3100	1.01
P2	3480	3500	1.01
P3[b)]	4600	4600	1.01
P4	6800	6900	1.02
P5	8900	9700	1.08
P6[c)]	13 400	14 600	1.07
P7	18 000	19 500	1.08
P8	26 400	30 200	1.11
P9	33 200	39 700	1.14
P10	36 100	44 500	1.17
P11	39 400	53 400	1.24
P12	57 800	71 700	1.12
P13	60 700	86 700	1.19

a)RMN ^{1}H: M_n = 3200 g mol^{-1}; MALDI-TOF-MS: M_n = 3300 g mol^{-1}; M_w = 3500 g mol^{-1}; *Đ* = 1.06; b)RMN ^{1}H: M_n = 4900 g mol^{-1}; MALDI-TOF-MS: M_n = 4500 g mol^{-1}; M_w = 4800 g mol^{-1}; *Đ* = 1.06; c)RMN ^{1}H: M_n = 12 600 g mol^{-1}.

Thus, the narrowly distributed PNAM samples (*Đ* < 1.2) with absolute M_w values ranging from 3000 to 87 000 g mol^{-1} could be considered as standard samples.

3.2. Optimization of ^{1}H DOSY NMR Measurements

3.2.1. Determination of the Critical Concentration of Chain Entanglement C*

^{1}H DOSY NMR measurements on polymer solutions have to be conducted in diluted conditions below the critical concentration of the polymer chain entanglement C*. Since C* decreases with the polymer MW, we thus determined C* for the *t*Bu-PNAM-DB sample **P12** with M_w = 71 700 g mol^{-1} in order to define the concentration not to exceed (Figure 2).

The average diffusion coefficients D_{PNAM} were measured by ^{1}H DOSY NMR both in D_2O and $CDCl_3$, two common solvents of PNAM, for various polymer concentrations (*C*) ranging from 0.625 to 60 mg mL^{-1}. In D_2O, each measurement was repeated five times for each concentration to estimate the standard deviation ΔD. The D_{PNAM} versus *C* plot indicated that D_{PNAM} was constant at low concentration (38 µm^2 s^{-1}). In contrast, D_{PNAM} values significantly decreased at high concentration because of polymer chain entanglement. The inflection point of the curve at 12–15 mg mL^{-1} was assimilated to C*. At the same time, it was observed that the repeatability of the D_{PNAM} measurements was lower for highly diluted solutions than for the concentrated ones, due to the corresponding lower signal-to-noise ratio. A compromise should thus be made for these measurements: sample

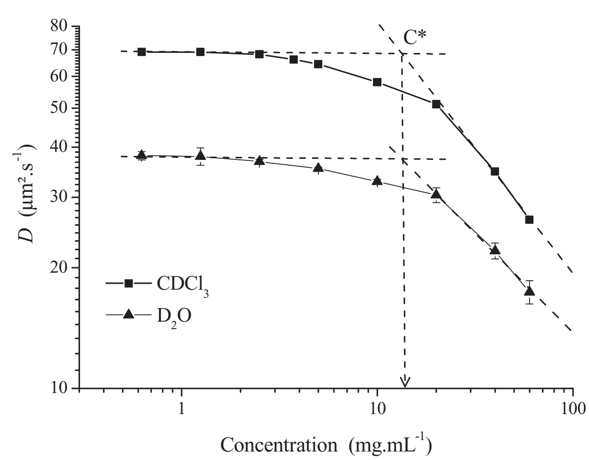

Figure 2. Variation of the average diffusion coefficient D_{PNAM} at 300 K with the concentration of *t*Bu-PNAM-DB homopolymer **P12** (M_w = 71 700 g mol^{-1}) in D_2O and $CDCl_3$. The critical concentration of the polymer chain entanglement C* was estimated as the concentration at the inflection point of the curve.

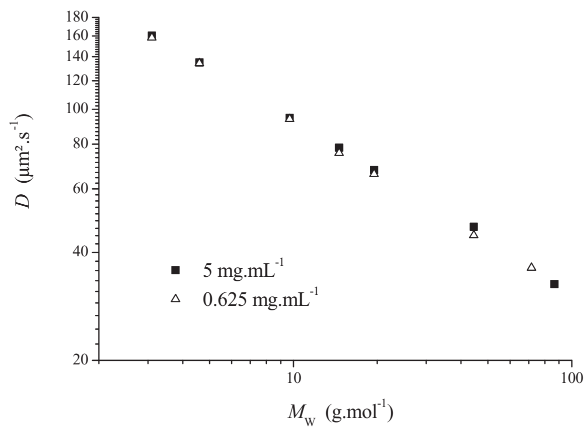

Figure 3. Variation of D_{PNAM} measured by ^{1}H DOSY NMR in D_2O at 300 K with the weight-average molecular weight (M_w) of PNAM samples determined by SEC/MALLS. Data points were acquired at two different PNAM concentrations ■ 5 mg mL^{-1} (Log $D = -0.4741$ Log $M_w + 3.8654$; $r^2 = 0.9992$) and △ 0.625 mg mL^{-1} (Log $D = -0.4759$ Log $M_w + 3.8648$; $r^2 = 0.9995$).

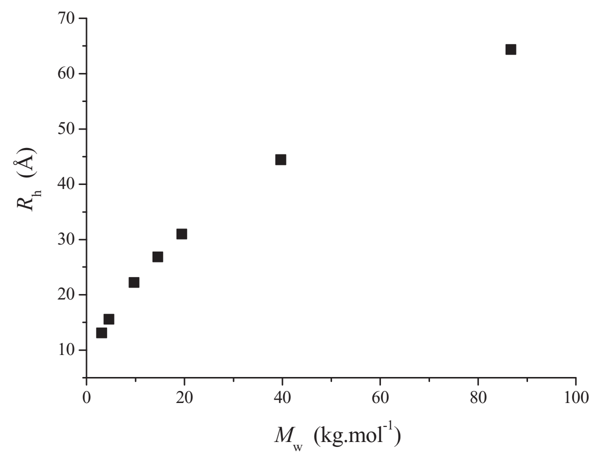

Figure 4. Variation of the average hydrodynamic radii (R_h) estimated from Equation (5) with the weight-average molecular weight (M_w) of PNAM samples. PNAM concentration in D_2O = 0.625 mg mL^{-1}. The viscosity η of the solutions was assumed to be the one of D_2O at $T = 300$ K (1.047×10^{-3} kg m^{-1} s^{-1}).[52]

concentration should be low enough to ensure that C < C* and not to affect viscosity but sufficiently high to provide precise values of D_{PNAM}.

In $CDCl_3$, D_{PNAM} values were higher than in D_2O (69 µm^2 s^{-1} at low concentration) due to the lower solvent viscosity. The measured value of C* was however similar (12–15 mg mL^{-1}).

3.2.2. Determination of the Solution Properties for PNAM in D_2O at 300 K at Different Polymer Concentrations

The variation of D_{PNAM} with the weight-average molecular weight (M_w) of several *t*Bu-PNAM-DB samples covering a range between 3 and 87 kg mol^{-1} was then assessed in D_2O at two different polymer concentrations below C*, 0.625 and 5 mg mL^{-1}, respectively. At least four measurements were repeated for each point (Figure 3).

Similar highly linear relationships were found between Log D_{PNAM} and Log M_w for the two polymer concentrations. Correlation coefficients r^2 were higher than 0.999 and the fitting equations were almost identical ($y = -0.475x + 3.865$). It indicated that, within this range, the polymer concentration had a negligible influence on the solution viscosity and that the latter can be assumed to be equal to the solvent viscosity. The exponent α for *t*Bu-PNAM-DB in D_2O at 300 K is represented by the absolute value of the slope of the Log D_{PNAM} versus Log M_w plot, i.e., $\alpha = 0.475$. As explained in the Introduction section, this α value can be viewed as the Flory exponent δ (fractal dimension, $d_F = 1/\delta = 2.1$) that is representative of a polymer in a θ solvent in this case.

$$R_h = \frac{k_B \times T}{6\pi \times \eta \times D} \quad (5)$$

The Stokes–Einstein equation (Equation (5) with the Boltzmann constant k_B) was used to estimate the average hydrodynamic radii, R_h, of the PNAM chains in D_2O from the D_{PNAM} values obtained at the higher dilution (0.625 mg mL^{-1}), by assuming that the viscosity η of the solutions was the one of pure D_2O at $T = 300$ K (1.047×10^{-3} kg m^{-1} s^{-1}).[52] R_h corresponds to the radius of an equivalent sphere that would exhibit the same diffusion than the polymer coil. As shown on Figure 4, the estimated R_h values increased from 13 to 64 Å (hydrodynamic diameters from 2.6 to 12.8 nm) for the *t*Bu-PNAM-DB samples with M_w ranging from 3 to 87 kg mol^{-1}. These values are similar to the one obtained by small angle neutron scattering (SANS)[53] and dynamic light scattering (DLS)[54] for PEG in the same solvent.

As already noticed, the standard deviations for the average diffusion coefficients were higher for the polymer solutions at 0.625 mg mL^{-1} than at 5 mg mL^{-1}. As an illustration, the corresponding M_w values (obtained from the above-mentioned fitting equation) deviated up to 12% from the ones determined by SEC/MALLS, whereas the maximum deviation was only 2% for the polymer solutions at 5 mg mL^{-1}. Therefore, we chose to perform the D_{PNAM} measurements at 5 mg mL^{-1} polymer concentration for the rest of this study.

3.3. Validation of the ^{1}H DOSY NMR Calibration Curve for the Determination of PNAM Molecular Weights in D_2O

The power law correlation between the diffusion coefficient and the molecular weight of the polymers thus makes ^{1}H DOSY NMR an interesting complementary method suitable for the determination of their molecular weight.[42,47] Indeed, the above-determined Log D_{PNAM} versus Log M_w plot can be used as a calibration curve to determine the M_w values of various types of PNAM samples from their D_{PNAM} measured by DOSY NMR.

To test the reliability of the method, the D_{PNAM} values were subsequently measured for other *t*Bu-PNAM-DB samples in an independent experiment series. The M_w values calculated from the previously determined calibration curve (Log D_{PNAM} = –0.475 Log M_w + 3.865) were compared to the values determined by SEC/MALLS. The deviation was generally below 5% and did not exceed 10%. This good agreement thus validated the method. Its robustness was further confirmed when the calibration curve was refined by incorporating the new data points (Figure 4). All the points were very well aligned and the correlation coefficient (r^2 = 0.999) remained high. The refined fitting equation was (y = –0.4789x + 3.883) and can be used as a new calibration curve to determine the molecular weights of unknown PNAM samples.

Table 2. Calibration curves, Flory exponent δ, and fractal dimension d_F determined for PNAM in various solvents.

Solvent	Standard equation	δ a)	d_F b)
D_2O	$M_w = 1.283 \times 10^8 D_{PNAM}{}^{-2.088}$	0.48	2.09
$CDCl_3$	$M_w = 5.154 \times 10^8 D_{PNAM}{}^{-2.096}$	0.48	2.10
PBS, pH 7.4	$M_w = 1.754 \times 10^8 D_{PNAM}{}^{-2.223}$	0.45	2.22
D_2O/MeOD 80/20	$M_w = 5.939 \times 10^8 D_{PNAM}{}^{-2.464}$	0.41	2.46

a)The Flory exponent δ was assimilated to the exponent α of the power law correlating D and M_w values; b)d_F is the fractal dimension determined from Equation (3).

3.4. Determination of Calibration Curves and Solution Properties of PNAM in Various Solvents

The same methodology as developed in D_2O was followed to determine the calibration curves and to estimate the Flory exponents for PNAM in various aqueous solvents and in chloroform (Figure 5 and Table 2).

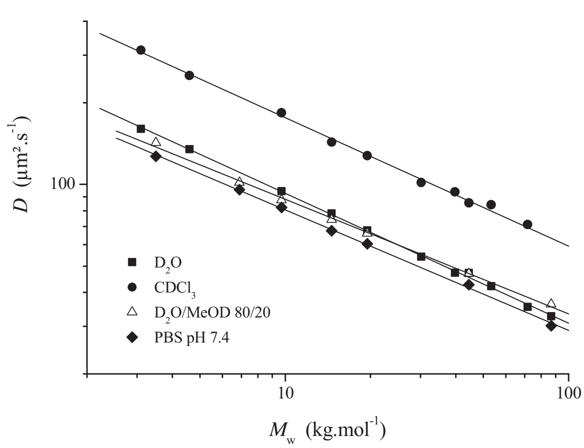

Figure 5. Influence of the solvent on the variation of D_{PNAM} with the weight-average molecular weight (M_w) of PNAM samples determined by SEC/MALLS. ■ D_2O (Log D = –0.4789 Log M_w + 3.883; r^2 = 0.999); ● $CDCl_3$ (Log D = –0.477 Log M_w + 4.1557; r^2 = 0.9963); △ D_2O/MeOD 80/20 v/v mixture (Log D = –0.4057 Log M_w + 3.5595; r^2 = 0.998); and ▼ PBS pH 7.4 (Log D = –0.4499 Log M_w + 3.709; r^2 = 0.9988). Polymer concentration = 5 mg mL⁻¹; T = 300 K.

As already noticed in Figure 2, D_{PNAM} values were significantly higher in $CDCl_3$ than in the aqueous solvents due to its lower viscosity. However, the Flory coefficient in the organic solvent (δ = 0.48) was identical to the one determined in D_2O, indicating that chloroform is also a θ solvent for PNAM.

The solution properties of PNAM were also studied in phosphate buffer (PBS, pH 7.4), a buffer regularly used for biological applications that is characterized by a moderate ionic strength representative of the physiological media. D_{PNAM} values and the Flory coefficient (δ = 0.45) were slightly lower than in pure D_2O, reflecting, respectively, an influence of the higher viscosity and of the increased ionic strength of the solvent.

To further validate the technique, the influence of a less good solvent was then explored. Methanol that is miscible to water and qualitatively exhibits poorer solvent properties for PNAM was chosen. Measurements were performed on clear solutions of *t*Bu-PNAM-DB in an 80/20 vol/vol D_2O/MeOD mixture. In this case, the Flory exponent was indeed lower (δ = 0.41), clearly indicating a lower solvent quality.

4. Conclusions

¹H DOSY NMR provides a simple and convenient method to determine both the molecular weight of PNAM samples and the solution properties of the PNAM chains in various organic and aqueous solvents. ¹H DOSY NMR was used to measure the average diffusion coefficients D_{PNAM} of a library of PNAM samples of increasing M_w ranging from 3 to 87 kg mol^{-1}, synthesized by RAFT polymerization. The estimated hydrodynamic radii of the corresponding PNAM chains, R_h, ranged from 13 to 64 Å in D_2O.

In all solvents, highly linear relationships were found between Log D_{PNAM} and Log M_w. Therefore, calibration curves were established and they could be used in the future for the reliable monitoring of the M_w values of various types of PNAM samples, such as raw polymerization samples or purified samples, highly diluted in the NMR solvent, even if they are available in small amount.

In addition, the slope of each calibration curve can be considered as the Flory exponent δ characterizing the solvent–polymer interaction. The measurements showed that D_2O and $CDCl_3$ are theta solvents for PNAM ($\delta = 0.48$). Compared to pure D_2O, the Flory exponent decreased slightly in a phosphate buffer at pH 7.4 (δ = 0.45), but more in a D_2O/MeOD 80/20 vol/vol mixture, confirming that methanol is not a good solvent of PNAM.

DOSY NMR thus appears as a valuable complementary technique to other characterization techniques such as SEC, NMR, and MALDI-TOF-MS, especially when the latter shows some limitations. For instance, when the polymer or its chain ends promote a strong adsorption on SEC columns or when the polymer molecular weight is too high to be determined either by ^{1}H NMR or MALDI-TOF-MS. DOSY NMR is also an alternative to viscosimetric and light scattering-based methodologies to study the solution properties of various polymers for which data are sparse.

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[1] J. Parrod, C. R. Jelles, *C. R. Acad. Sci.* **1956**, *243*, 1040.
[2] G. E. Ham*(Monsanto Chemical Co.), US 2,683,703; Chem. Abstr.* **1954**, *48*, 1184.
[3] J. Elles, *Chim. Med.* **1959**, *4*, 26.
[4] P. R. Thomas, D. J. Attfield*(British Nylon Spinners Ltd.), Brit. 914,780;* **1964**, *58*, P5810c.
[5] D. Lim, J. Kopeck, H. Bazhilova, J. Vacik, *(Ceskoslovenka Akademie Ved), Ger. Offen. 2,151,909,27; Chem. Abstr.* **1972**, *77*, 49139f.
[6] M. Gorman, Y. H. Chim, A. Hart, M. O. Riehle, A. J. Urquhart, *J. Biomed. Mater. Res. Part A* **2014**, *102*, 1809.
[7] O. Schiavon, P. Caliceti, P. Ferruti, F. M. Veronese, *Farmaco* **2000**, *55*, 264.
[8] C. K. Narang, K. Brunfeldt, K. E. Norris, *Tetrahedron Lett.* **1977**, *18*, 1819.
[9] R. Epton, P. Goddard, G. Marr, J. V. McLaren, G. J. Morgan, *Polymer* **1979**, *20*, 1444.
[10] R. Epton, S. J. Hocart, G. Marr, *Polymer* **1980**, *21*, 481.
[11] K. Ishii, Z. Honda (Daicel Ltd.), *Ger. Offen. 2,414,795,27; Chem. Abstr.* **1973**, *82*, 58534.
[12] R. Epton, C. Holloway, J. V. McLaren, *J. Chromatogr. A* **1974**, *90*, 249.
[13] R. Epton, S. R. Holding, J. V. McLaren, *J. Chromatogr. A* **1975**, *110*, 327.
[14] C. Gelfi, P. de Besi, A. Alloni, P. G. Righetti, *J. Chromatogr. A* **1992**, *608*, 333.
[15] V. P. Torchilin, V. S. Trubetskoy, K. R. Whiteman, P. Caliceti, P. Ferruti, F. M. Veronese, *J. Pharm. Sci.* **1995**, *84*, 1049.
[16] F. M. Veronese, O. Schiavon, P. Calceti, L. Sartore, E. Ranucci, P. Ferruti(Consiglio Nationale delle Ricerche, Italy), *U.S. 5,629,384*, **1997**.
[17] A. Favier, M. T. Charreyre, P. Chaumont, C. Pichot, *Macromolecules* **2002**, *35*, 8271.
[18] A. Favier, M. T. Charreyre, C. Pichot, *Polymer* **2004**, *45*, 8661.
[19] Y. S. Jo, A. J. van der Vlies, J. Gantz, S. Antonijevic, D. Demurtas, D. Velluto, J. A. Hubbell, *Macromolecules* **2008**, *41*, 1140.
[20] L. Albertin, A. Wolnik, A. Ghadban, F. Dubreuil, *Macromol. Chem. Phys.* **2012**, *213*, 1768.
[21] A. Favier, F. D'Agosto, M. T. Charreyre, C. Pichot, *Polymer* **2004**, *45*, 7821.
[22] G. Gody, P. Boullanger, C. Ladavière, M.-T. Charreyre, T. Delair, *Macromol. Rapid Commun.* **2008**, *29*, 511.
[23] J. Chiefari, Y. K. Chong, F. Ercole, J. Krstina, J. Jeffery, T. P. T. Le, R. T. A. Mayadunne, G. F. Meijs, C. L. Moad, G. Moad, E. Rizzardo, S. H. Thang, *Macromolecules* **1998**, *31*, 5559.
[24] A. Favier, C. Ladaviere, M. T. Charreyre, C. Pichot, *Macromolecules* **2004**, *37*, 2026.
[25] A. Favier, M. T. Charreyre (bioMérieux S.A.), *PCT Int. Appl. WO 04/055060*, **2004**.
[26] B. de Lambert, M.-T. Charreyre, C. Chaix, C. Pichot, *Polymer* **2007**, *48*, 437.
[27] M. Bathfield, F. D'Agosto, R. Spitz, M.-T. Charreyre, C. Pichot, T. Delair, *Macromol. Rapid Commun.* **2007**, *28*, 1540.
[28] M. Bathfield, D. Daviot, F. D'Agosto, R. Spitz, C. Ladaviere, M.-T. Charreyre, T. Delair, *Macromolecules* **2008**, *41*, 8346.
[29] S. Adjili, A. Favier, J. Massin, Y. Bretonniere, W. Lacour, Y.-C. Lin, E. Chatre, C. Place, C. Favard, D. Muriaux, C. Andraud, M.-T. Charreyre, *RSC Adv.* **2014**, *4*, 15569.
[30] B. de Lambert, C. Chaix, M.-T. Charreyre, A. Laurent, A. Aigoui, A. Perrin-Rubens, C. Pichot, *Bioconjugate Chem.* **2005**, *16*, 265.
[31] C. Cepraga, T. Gallavardin, S. Marotte, P.-H. Lanoe, J.-C. Mulatier, F. Lerouge, S. Parola, M. Lindgren, P. L. Baldeck, J. Marvel, O. Maury, C. Monnereau, A. Favier, C. Andraud, Y. Leverrier, M.-T. Charreyre, *Polym. Chem.* **2013**, *4*, 61.
[32] A. Favier, *PhD Thesis*, University of Lyon, **2002**.
[33] M.-T. Charreyre, F. D'Agosto, A. Favier, C. Pichot, B. Mandrand (bioMérieux, S.A.), *PCT Int. Appl. WO 01/92361 A1*, **2001**.
[34] E. O. Stejskal, J. E. Tanner, *J. Chem. Phys.* **1965**, *42*, 288.
[35] K. F. Morris, C. S. Johnson, *J. Am. Chem. Soc.* **1992**, *114*, 3139.
[36] C. S. Johnson, *Prog. Nucl. Magn. Reson. Spectrosc.* **1999**, *34*, 203.
[37] H. Walderhaug, O. Söderman, D. Topgaard, *Prog. Nucl. Magn. Reson. Spectrosc.* **2010**, *56*, 406.
[38] A. Jerschow, N. Müller, *Macromolecules* **1998**, *31*, 6573.

[39] G. Fleischer, A. Puhlmann, F. Rittig, Č. Koňák, *Colloid Polym. Sci.* **1999**, *277*, 986.
[40] S. Viel, M. Mazarin, R. Giordanengo, T. N. T. Phan, L. Charles, S. Caldarelli, D. Bertin, *Anal. Chim. Acta* **2009**, *654*, 45.
[41] S. Augé, P.-O. Schmit, C. A. Crutchfield, M. T. Islam, D. J. Harris, E. Durand, M. Clemancey, A.-A. Quoineaud, J.-M. Lancelin, Y. Prigent, F. Taulelle, M.-A. Delsuc, *J. Phys. Chem. B* **2009**, *113*, 1914.
[42] S. Viel, D. Capitani, L. Mannina, A. Segre, *Biomacromolecules* **2003**, *4*, 1843.
[43] X. Liu, J. Wu, M. Yammine, J. Zhou, P. Posocco, S. Viel, C. Liu, F. Ziarelli, M. Fermeglia, S. Pricl, G. Victoreto, C. Nguyen, P. Erbacher, J.-P. Behr, L. Peng, *Bioconjugate Chem.* **2011**, *22*, 2461.
[44] C. Barrère, M. Mazarin, R. Giordanengo, T. N. T. Phan, A. Thévand, S. Viel, L. Charles, *Anal. Chem.* **2009**, *81*, 8054.
[45] O. Assemat, M.-A. Coutouly, R. Hajjar, M.-A. Delsuc, *C. R. Chim.* **2010**, *13*, 412.
[46] J. Viéville, M. Tanty, M. A. Delsuc, *J. Magn. Reson.* **2011**, *212*, 169.
[47] W. Li, H. Chung, C. Daeffler, J. A. Johnson, R. H. Grubbs, *Macromolecules* **2012**, *45*, 9595.
[48] M. L. Conrad, A. C. Moser, D. S. Hage, *J. Sep. Sci.* **2009**, *32*, 1145.
[49] D. Wu, A. Chen, C. S. Johnson, Jr., *J. Magn. Reson. A* **1995**, *115*, 264.
[50] M. A. Delsuc, T. E. Malliavin, *Anal. Chem.* **1998**, *70*, 2148.
[51] J.-L. Pons, T. E. Malliavin, M. A. Delsuc, *J. Biomol. NMR* **1996**, *8*, 445.
[52] B. Le Neindre, *Techniques de l'Ingénieur* **2007**, *K493*, p.1.
[53] M. L. Alessi, A. I. Norman, S. E. Knowlton, D. L. Ho, S. C. Greer, *Macromolecules* **2005**, *38*, 9333.
[54] K. L. Linegar, A. E. Adeniran, A. F. Kostko, M. A. Anisimov, *Colloid J.* **2010**, *72*, 279.